# Blitz text test bench

Hard cases for extraction and read-aloud. Every case has an ID like J-03.

How to use this document. Open it in Blitz and press play on any case. Section J holds things that should not be spoken, section S holds symbols that must be spoken meaningfully, section P holds prose mechanics, and section M holds mathematics in context. The companion report lists, for each case, what a good reader would say.

# J. Junk that should not be spoken

Each case below contains something a listener should never hear.

**J-01.** *Tall brackets drawn by the maths engine.*

A column vector $\begin{bmatrix} x \\ y \\ z \\ w \\ 1 \end{bmatrix}$ is written vertically on the page.

**J-02.** *Matrix bracket pieces set as separate glyphs.*

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

The grid above is the identity matrix, built the way older equation editors build it.

**J-03.** *Box-drawing characters in a directory tree.*

```
app/
├── renderer/
│   └── main.js
└── electron/
    └── tts.js
```

The tree shows where the two scripts live.

**J-04.** *Dot leaders in a table of contents.*

Chapter 1 Vectors and Scalars ..... 17
Chapter 2 The Calculus of Motion ..... 43
Chapter 3 Springs ..... 81

**J-05.** *Running header and page number.*

Every page of this document carries a running header (“Blitz text test bench”) and a page number in the footer. Neither should be read aloud, and neither should be glued onto the first or last sentence of a page.

SPECIMEN

SPECIMEN

**J-06.** *A sentence broken across a page.*
This sentence is placed at the very bottom of a page so that it has to continue on the next page, which means the page number and the running header sit in the middle of it when the text is extracted page by page, and a reader that segments one page at a time will cut it into two pieces, each of which it will then treat as a sentence of its own, with the first piece ending in mid-air and the second piece starting in the

middle of a clause that the listener has already half forgotten by the time the next page has been fetched, rendered and read.

**J-07.** *Private-use glyph (icon font).*
☞ Keep the lid closed while the centrifuge is spinning.

**J-08.** *Unmappable glyph arriving as a control character.*
The impulse equals the change in momentum, Δp, over the short interval Δt.

**J-09.** *Zero-width space inside a word.*
The database was rebuilt overnight, and the timestamp column was kept.

**J-10.** *Soft hyphen inside a word.*
Tune one hyperparameter at a time, and never more than one metaparameter.

**J-11.** *Decorative bullets.*
❖ Weigh the sample.
➤ Dissolve it in water.
■ Record the temperature.

**J-12.** *Labels inside a figure.*

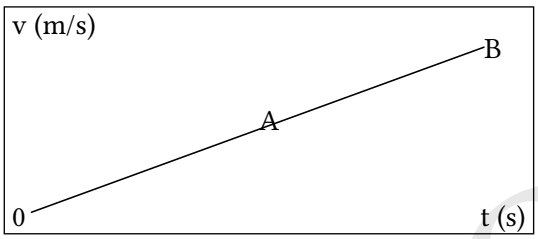



Figure 1: Velocity against time for a falling ball.

**J-13.** *Ornamental section break.*
The first scene ends here.

❧ ❧ ❧

The second scene begins the next morning.

**J-14.** *Line numbers in a code listing.*

```
let total = 0;
for (const x of xs) total += x;
return total;
```

The loop on line 2 adds every element.

**J-15.** *Watermark behind the text.*
This page carries a large diagonal SPECIMEN watermark behind the body text. It is decoration, not content.

SPECIMEN

# S. Symbols that must be spoken

Each case below contains a symbol whose meaning must survive into speech.

**S-01.** *Capital delta.*
The change Δx is small compared with x itself.

**S-02.** *Greek letters in prose.*
The angles α, β and θ sum to π, and λ is the wavelength while σ is the spread.

**S-03.** *Comparison signs.*
If x ≤ 5 and y ≥ 2, then x ≠ y need not hold, and π ≈ 3.14.

**S-04.** *Plus-minus, times, divide.*
The rod is 12 ± 0.5 cm long; 3 × 4 = 12 and 12 ÷ 3 = 4.

**S-05.** *Degrees.*
Water boils at 100 °C, the body sits near 37 °C, and a right angle is 90°.

**S-06.** *Percent and per mille.*
Prices rose 15% this year, and the salinity is 35‰.

**S-07.** *Currencies.*
It cost $5.99 here, €20 in Paris, £3 in London, ¥500 in Tokyo and ₹100 in Delhi.

**S-08.** *Micro sign and units.*
Each cell is 10 µm wide, the dose is 5 µg/dL, and g = 9.81 m/s².

**S-09.** *Vulgar fractions.*
Add ½ cup of flour; ¾ of the class passed and ⅓ stayed behind.

**S-10.** *Unicode superscripts and subscripts.*
We have x² + y³, a rate of $10^{-5}$ per second, and water is $H_2O$ while $CO_2$ is a gas.

**S-11.** *Typeset superscripts and subscripts.*
We have $x^2 + y^3$, a rate of $10^{-5}$ per second, and water is $H_2O$ while $CO_2$ is a gas.

**S-12.** *Arrows.*
The reaction A ⟶ B is fast, the rule p ⟹ q is a premise, and prices went ↑ while sales went ↓.

**S-13.** *Primes.*
Compare f′(x) with f″(x); the shelf is 5′ 11″ tall.

**S-14.** *Big operators as text.*
The Σ of the terms diverges, the ∫ from 0 to ∞ converges, and √2 is irrational.

**S-15.** *Set and logic symbols.*
For every x ∈ S there is a y ∉ T; ∀ε > 0 ∃δ > 0 such that the bound holds.

**S-16.** *Mathematical italic letters as text.*
The probability $P(A)$ rises with the parameter $\theta$, and $n$ is the sample size.

**S-17.** *Defined-as and other relations.*
Let $f \triangleq g + 1$, assume a ∝ b, and note that A ⊆ B.

**S-18.** *Section, copyright and trademark signs.*
See § 4.2 and ¶ 3; the text is © 2026 Blitz, and Kokoro™ is named here only as an example®.

**S-19.** *Ellipsis.*
Wait… then what happened to the other half?

**S-20.** *En dash ranges and pairs.*
Read pages 10–20 of the 1990–1995 report on the London–Paris line.

**S-21.** *Em dash.*
The result—surprisingly—was exactly zero.

**S-22.** *Minus sign versus hyphen.*
At −40 °C the two scales agree; the hyphenated -40 is the same number.

**S-23.** *Ampersand, hash, slash and ratio.*
R&D spent #3 of the budget on a 3:2 ratio of km/h tests and/or audits.

# P. Prose mechanics

Each case below exercises the way ordinary prose is typeset.

**P-01.** *Ligatures fi fl ff ffi ffl.*
The official efficient affluent fluffy fjord shuffle baffled five office staff.

**P-02.** *Hyphenated line ends.*

Internationalisation and incomprehensibility are characteristically trouble-some for straightforward extraction.

**P-03.** *Compound hyphen at a line end.*

A well-known self-contained state-of-the-art object-oriented design.

**P-04.** *Tight letter tracking.*
Tracking minus five hundredths keeps these words apart.
Tracking minus twelve hundredths may start to weld.
Tracking minus two tenths welds these words together.

**P-05.** *Wide letter-spacing.*
W I D E L Y  S P A C E D  H E A D I N G
The heading above is letter-spaced the way chapter openers often are.

**P-06.** *Small caps.*
Nasa and the Royal Society both run Unix servers.

**P-07.** *Drop cap.*

Once upon a time a reader met a drop cap and had to decide what to make of it. The large letter sits in its own box beside the paragraph.

**P-08.** *Footnote markers.*
This claim[1] needs support, and so does this one[2].

**P-09.** *URL.*
Visit https://example.org/docs/v2?lang=en for the details.

**P-10.** *Email address.*
Write to reader.help@example.com with questions.

[1] The note text appears at the foot of the page.
[2] A second note.

**P-11.** *Code identifiers in prose.*
Call `parse_config()` first, then `foo.bar(x)`, and finally `getElementById` on the `MAX_SIZE` buffer.

**P-12.** *Numbers.*
The total was 1,234.56 units; version 3.10 replaced 3.9; the tolerance is 1e-5; and 0.000001 is a millionth.

**P-13.** *Dates and times.*
The run started on 2026-10-24 at 14:30 and ended on 24 Oct 2026 at 4:05 pm.

**P-14.** *Abbreviations with full stops.*
Some solvents, e.g. ethanol, are safe, i.e. mostly. Dr. Rao showed in Fig. 3 that approx. 4 km is enough, as Smith et al. (2019) found on p. 12, vs. 6 km before.

**P-15.** *Nested quotes and parentheses.*
“She said, ‘the (very (deeply) nested) case’ works,” he wrote (and meant it).

**P-16.** *Numbered list.*
1. Heat the solution.
2. Add the reagent.
3. Stir for two minutes.

**P-17.** *Lettered list in a sentence.*
The kit needs (a) a beaker, (b) a burner and (c) a timer.

**P-18.** *Two-column section.*

The left column starts this thought and carries it down to the bottom of the column before it stops.

The right column picks the story up again and finishes it here.

**P-19.** *Heading without a full stop.*

## Energy and Work

Work is force times distance.

**P-20.** *Roman numerals.*
Chapter IV covers Henry VIII and the aftermath of World War II.

**P-21.** *Accented letters.*
The naïve café owner admired the façade designed by Schrödinger’s cousin.

**P-22.** *Initials and acronyms.*
J. R. R. Tolkien never visited the U.S.A., but NASA and the SQL committee did.

**P-23.** *Citations.*
Earlier work [12, 14] disagreed with Smith (2019) and with the review by Lee et al. [7].

**P-24.** *A small table.*

| Element | Symbol | Mass (u) |
|---|---|---|
| Hydrogen | H | 1.008 |
| Carbon | C | 12.011 |

**P-25.** *Code block.*

```
for (int i = 0; i < n; i++) {
  sum += a[i];
}
```

The loop above sums the array.

**P-26.** *Chemical charges.*
Iron appears as $Fe^{3+}$ and sulfate as $SO_4^{2-}$ in the solution.

# M. Mathematics in context

Each case below puts mathematics where a reader meets it.

**M-01.** *Inline maths mid-sentence.*
The energy $E = mc^2$ follows from the rest mass alone.

**M-02.** *Inline fraction.*
The ratio $\frac{a}{b}$ equals $\frac{1}{2}$ when $b = 2a$.

**M-03.** *Numbered display equation.*
Newton gives the law

$$F = ma \tag{1}$$

which we use throughout.

**M-04.** *Display with a limit and fraction.*
The velocity is

$$v = \lim_{\Delta t \to 0} \frac{\Delta p}{\Delta t}$$

which is the derivative of position.

**M-05.** *Matrix.*
The rotation matrix is

$$R = \begin{pmatrix} \cos\theta & -\sin\theta \\ \sin\theta & \cos\theta \end{pmatrix}$$

for an angle $\theta$.

**M-06.** *A sentence that is only maths.*
$x^2 + y^2 = r^2$.

**M-07.** *Sum and integral displays.*

$$\sum_{i=1}^{n} i = \frac{n(n+1)}{2} \quad \text{and} \quad \int_0^1 x^2 \, \mathrm{d}x = \frac{1}{3}$$

**M-08.** *Sub- and superscripts inline.*
Each term $x_i^2$ multiplies the entry $a_{ij}$ of the matrix.

**M-09.** *Greek inside maths.*
We require $\alpha + \beta = \gamma$ and $\Delta x > 0$.

**M-10.** *Roots, norms and absolute values.*
The length is $\sqrt{x^2 + y^2}$, written $\|\boldsymbol{v}\|$ or $|v|$.

**M-11.** *Vectors with arrows and hats.*
The dot product $\vec{v} \cdot \vec{w}$ uses the unit vector $\hat{x}$.

**M-12.** *Set-builder notation.*
The positive reals are $\{x \in \mathbb{R} \mid x > 0\}$.

**M-13.** *Piecewise definition.*

$$f(x) = \begin{cases} x \text{ if } x \geq 0 \\ -x \text{ otherwise} \end{cases}$$

This is the absolute value.